

FORT ST JAMES, BC, Canada

WMO: 719330

Lat: 54.455N

Lon: 124.286W

Elev: 2249

StdP: 13.54

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.1	-33.0	-46.3	0.4	-38.0	-41.5	0.5	-33.2	18.6	4.1	16.7	1.2	2.8	50	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	82.0	63.4	78.7	62.2	75.4	60.8	66.0	76.8	64.4	74.7	62.8	72.3	9.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	92.2	70.1	60.8	86.4	68.4	59.0	80.9	67.1	32.1	76.8	30.8	74.1	29.5	72.2	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.5	14.4	-44.4	87.7	4.5	3.0	-47.6	89.9	-50.2	91.6	-52.7	93.3	-55.9	95.4		
(5)			-44.4	69.0	4.5	2.2	-47.6	70.6	-50.2	71.8	-52.7	73.1	-55.9	74.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.4	-4.9	-0.7	11.4	30.7	46.1	57.9	62.8	59.0	48.8	35.0	15.3	1.0		
	DBStd	26.68	16.59	14.43	16.29	12.24	9.66	7.07	5.68	6.56	7.93	8.40	12.59	15.65		
	HDD50	8257	1701	1421	1196	582	188	17	0	10	114	469	1041	1518		
	HDD65	12717	2166	1841	1661	1029	588	229	109	204	486	930	1491	1983		
	CDD50	1093	0	0	0	3	67	253	399	288	79	4	0	0		
	CDD65	77	0	0	0	0	2	15	42	17	1	0	0	0		
	CDH74	989	0	0	0	0	44	221	489	216	19	0	0	0		
	CDH80	172	0	0	0	0	5	37	91	37	2	0	0	0		
	WSAvg	6.4	5.7	5.9	6.5	7.5	7.8	7.1	6.5	6.2	6.3	6.2	5.8	5.6		
	PrecAvg	18.9	1.9	1.1	1.1	0.8	1.5	1.8	1.9	1.9	1.8	1.7	1.6	1.9		
(15) Precipitation	PrecMax	25.2	5.0	3.0	2.4	2.0	4.2	4.4	4.9	4.1	3.8	3.0	3.0	4.1		
	PrecMin	13.7	0.6	0.2	0.2	0.2	0.2	0.3	0.3	0.2	0.4	0.7	0.0	0.5		
	PrecStd	3.2	1.0	0.6	0.5	0.5	0.9	1.0	1.1	1.1	0.9	0.6	0.6	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	37.6	48.5	64.8	79.1	84.3	86.4	84.2	76.8	62.5	46.1	37.2		
		MCWB	32.4	33.9	38.4	48.3	56.9	63.2	65.9	65.9	61.3	51.1	39.9	32.9		
	2%	DB	29.8	30.2	42.9	58.0	73.4	79.2	82.4	79.2	70.1	55.5	39.1	31.6		
		MCWB	27.7	27.2	35.4	44.5	53.9	60.6	63.9	63.8	57.7	47.1	35.1	29.5		
	5%	DB	24.2	24.9	39.0	53.2	68.1	75.8	79.1	75.5	66.1	51.0	35.1	26.7		
		MCWB	23.0	22.8	33.3	42.0	52.0	59.3	63.1	62.0	55.8	44.0	32.8	25.1		
	10%	DB	17.7	19.8	34.2	48.3	63.6	72.9	76.3	72.0	62.4	46.8	32.0	22.6		
		MCWB	16.8	18.3	30.2	39.6	49.7	58.5	62.2	60.5	53.8	42.2	30.5	21.5		
	0.4%	WB	32.7	34.2	39.5	49.3	59.4	66.2	68.8	68.7	62.4	52.8	40.3	33.3		
		MCDB	34.7	37.7	47.4	62.6	74.3	78.8	81.1	79.5	73.2	60.5	44.8	36.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.7	27.3	36.1	45.2	55.8	63.7	66.5	65.6	59.2	47.7	35.5	28.9		
		MCDB	29.6	29.5	41.8	56.3	69.9	74.6	77.3	75.6	67.8	53.8	38.0	30.7		
	5%	WB	22.4	22.8	33.4	42.4	53.2	61.7	65.0	63.6	56.8	44.8	32.6	24.9		
		MCDB	23.8	24.9	38.4	51.9	65.9	72.0	75.4	72.9	63.8	49.7	34.3	26.3		
	10%	WB	16.7	18.5	30.3	39.7	50.6	59.7	63.5	61.8	55.0	42.4	30.2	20.8		
		MCDB	17.7	20.1	34.6	47.9	61.9	69.7	73.3	70.1	61.0	46.4	31.8	22.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	21.1	23.8	22.3	23.0	22.9	22.0	21.4	19.9	15.2	15.3	16.7		
		MCDBR	22.7	23.3	24.3	27.5	30.4	27.8	26.7	26.7	25.6	21.6	18.1	18.9		
		MCWBR	20.8	20.3	17.8	16.7	15.0	13.3	12.6	13.7	14.6	14.2	14.5	17.0		
		MCDBR	22.2	23.1	24.0	26.3	28.5	24.4	23.4	24.2	22.4	19.8	16.6	18.1		
	5% WB	MCWBR	20.6	20.7	18.1	16.5	15.0	12.7	12.1	13.2	13.5	13.8	14.3	16.7		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.266	0.282	0.317	0.333	0.348	0.336	0.338	0.305	0.293	0.274	0.234		
	taud		2.331	2.331	2.341	2.400	2.476	2.454	2.525	2.515	2.610	2.594	2.388	2.321		
	Ebn at Noon		233	266	287	290	290	285	287	280	279	258	218	213		
	Edh at Noon		17	24	30	32	32	33	30	29	23	19	16	14		
(44) All-Sky Solar Radiation	RadAvg		211	501	919	1428	1717	1871	1809	1534	1019	539	237	159		
	RadStd		25	61	72	71	126	153	166	72	87	60	30	18		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+3.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A	

Nomenclature: See separate page